PROJECT GUIDE: AUTONOMOUS SOLAR CHARGER (MVP – PHASE 1)

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Area: Cyber-Physical Engineering / Autonomous Infrastructure

Project Focus: Intelligent Off-Grid Renewable Energy System

Status: Phase 1 (Bench Prototype) – Completed

1. Project Overview & Evolutionary Roadmap

This project develops a modular off-grid solar energy infrastructure isolated from the traditional electrical grid. Phase 1 validates a functional portable charger capable of harvesting solar energy, managing lithium-ion battery storage safely, and outputting a regulated 5.0 V USB rail. This prototype serves as a baseline node scaling toward microgrid edge monitoring and autonomous tracking systems.

The Scalability Ecosystem

- **Phase 1 (Current - Bench Harvesting MVP):** 6V / 5W PV panel, 18650 cell, TP4056 CC/CV control module, 5.0 V Boost converter output.
- **Phase 2 (Active - Automation & Tracking):** Quad-LDR differential sensing array with dual-axis servo tracking operating on a 10 Hz control loop.
- **Phase 3 (Scaling - Power Bank Infrastructure):** 30W / 18V PV input, 3S2P Li-ion bank (11.1V nominal / 12.6V peak), active BMS, and AC inverter integration.

2. System Architecture & Circuit Engineering Decisions

Functional Flow Diagram

[HARVESTING: 6V/5W Panel] --> [SAFETY: TP4056 Manager] --> [STORAGE: 18650 (3.7V)] --> [REGULATION: Boost Converter] --> [OUTPUT: 5V USB]

Technical Justification of Circuit Parameters

- **Overhead Solar Input (6V Nominal vs. 5V Direct):** Photovoltaic output fluctuates dynamically with irradiance. Charging a Li-ion cell to its 4.2 V cutoff requires input voltage overhead to overcome protection diode drops and thermal losses under partial cloud cover.
- **Charge Management Protocol (TP4056):** Li-ion cells are sensitive to thermal and electrical stress. The TP4056 implements constant-current / constant-voltage (**CC/CV**) profiles: delivering maximum initial current, transitioning to voltage clamping at 4.2 V, and cutting off power fully upon saturation.
- **Electrochemical Storage (18650 Cell):** Industrial standard 3.7 V nominal (2500 mAh) battery acting as an energy buffer, enabling continuous telemetry and load delivery independent of instantaneous solar flux.
- **High-Efficiency Switching Boost Converter (3.0 V–4.2 V to 5.0 V DC):** Battery cell output swings from 3.0 V (depleted) to 4.2 V (full). Inductor-based magnetic switching raises and stabilizes output to a fixed 5.0 V ± 0.15 V, avoiding heat dissipation common in linear regulators.

3. Bill of Materials (BOM) & Bench Infrastructure

Component / Module	Specification / Model	Qty / Rating	System Function
Photovoltaic Panel	Polycrystalline Glass Panel	6 V / 5 W	Solar irradiance harvesting source
Charge Controller	TP4056 with Protection Chip	1 A CC/CV	Battery charge regulation & cutoff
Lithium Storage Cell	18650 Li-ion Battery	3.7 V / 2500 mAh	Electrochemical energy

			accumulator
DC-DC Boost Converter	Step-Up Inductive Regulator	5.0 V Fixed USB	Regulated load power bus
Cell Holder	Molded ABS Plastic Frame	Single 18650 Slot	Connection without thermal battery stress

Bench Equipment	**Technical Class**	**Set Temp / Value**	**Bench Utility**
Soldering Station	Hikari HK-936B ESD-Safe	320 °C Target	Precision connection & wire tinning
Digital Multimeter	Minipa ET-1002 True RMS	CAT II 600 V	Voltage, current & short-circuit audit
Workbench Work Mat	Antistatic Thermal Silicone	500 °C Resistance	Thermal protection & SMD organization

4. Assembly Protocol & Pre-Commissioning Safety Audit

Step 1: Preparation & Thermal Tinning Protocol

1. Set soldering iron temperature to 320 °C. Clean tip on brass wool and tin with a 3 mm solder layer for optimal heat transfer.
2. Strip approximately 3 mm of insulation from 22 AWG stranded copper wiring.
3. **Pre-Tinning Requirement:** Apply flux and a thin coat of solder to stripped wire ends and PCB pads before mechanical connection to prevent cold joints.

Step 2: Circuit Interconnection

1. **Battery Rail:** Solder 18650 holder leads directly to B+ (red/positive) and B- (black/negative) pads of the TP4056 module.
2. **Output Rail:** Connect OUT+ on TP4056 to IN+ on the Boost module; connect OUT- to IN-.
3. **Generation Rail:** Connect solar panel positive lead to IN+ on TP4056 and negative lead to IN-.
4. **Dielectric Insulation:** Apply heat-shrink tubing across all exposed solder points to guarantee short-circuit isolation.

Step 3: Verification & Test Routine (Mandatory Pre-Power Check)

Prior to inserting the 18650 cell or connecting sensitive electronics, complete this audit:

- **Short-Circuit Continuity Test:** Place multimeter in Continuity Beep mode. Measure across positive and negative rails. *Zero sound must be emitted*. Any continuous beep indicates a solder bridge that must be rectified prior to cell insertion.
- **Solar Input Irradiance Test:** Expose panel to sunlight; verify input voltage at TP4056 IN+/IN- reads ≈ 6.0 V DC.
- **Regulated Bus Output Test:** Insert 18650 cell. Measure Boost USB output pins. Confirm output is stable between 4.9 V and 5.1 V DC.

5. Scalability Horizon & Microgrid Research Alignment

The architecture developed in Phase 1 establishes the baseline hardware node for advanced research vectors:

- **Autonomous Dual-Axis Solar Tracking (Phase 2):** Integrating microcontrollers (ESP32/Arduino) with quad-LDR sensors increases total daily photon harvesting efficiency by up to 40% compared to static arrays.
- **Off-Grid High-Power Conversion (Phase 3):** Scaling battery architecture to 3S/4S configurations and interfacing pure sine wave inverters (12 V DC -> 110 V / 220 V AC) transforms the system into a micro-power plant.
- **Cyber-Physical Edge Sensing (Dr. Goli Lab Alignment):** Deploying real-time serial/MQTT telemetry to monitor voltage sag, state-of-charge degradation, and sensor tampering (FDIA attacks) on microgrid nodes.

Status & Strategic Coordination Note

- **Mentor Review (Winder & Levi Alignment):** Phase 1 hardware execution and documentation structure fully validated by mentors.
- **Technical Execution Window:** Active alignment and review meetings are scheduled for **Friday**. Daily execution protocol operates on standard working hours (focusing on daytime research and bench testing).